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Compression artifacts reduction with multiscale tensor regularization

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Abstract
We study a multiscale tensor regularization based JPEG decompression artifact removal in digital images. Structure tensor eigenvalues based robust edge map is used within a variable exponent regularization. Variational constrained minimization which combines data fidelity driven by color subsampling and discrete cosine transformation operator is utilized. Experimental results across different compression levels and with various error metrics indicate our proposed method obtains high quality results on cartoon/clip-art and LIVE1 natural image databases.

Keywords JPEG deblocking · Artifact reduction · Edge preserving · Multiscale · Structure tensor · Regularization

1 Introduction

Joint Photographic Experts Group (JPEG) is a lossy compression widely used in digital image transmission and storage despite the introduction of better compression standard JPEG2000 (Wallace 1992; Rabbani 2002). Blocking artifacts are a common drawback in JPEG compressed images, and especially at low bit rates the artifacts manifest as Gibbs

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oscillations near edges due to the cancellation of discrete cosine transform (DCT). Due to this manifestation, compression methods can suffer from artifacts (Kulalvaimozhi et al. 2019). Over the years, various post processing techniques have been proposed to remove the blocking artifacts and they can be grouped into three major categories (Shen and Kuo 1998): (a) adaptive filtering which consider JPEG compression artifacts as noise (Yim and Ik Cho 2005; Xu et al. 2011; Pang et al. 2015; Zhang et al. 2014), (b) regularization methods which use prior smoothness constraints (Alter et al. 2005; Bredies and Holler 2012b), and (c) learning methods which approximate the compressed to uncompressed mapping (Dong et al. 2015), with learning methods performing better with added computational complexity compared to filters and regularization models.

Regularization methods can be adapted to improve deblocking by the way of using a strong prior functions with constraints from given compressed JPEG data. Among the regularization methods, the total variation (TV) (Rudin et al. 1992; Alter et al. 2005; Bredies and Holler 2012b; Prasath 2011; Prasath et al. 2017; Thanh et al. 2019) is used and provide better results than quadratic regularization terms but can still inherit staircasing artifacts. Recently, total generalized variation (TGV) (Bredies and Holler 2012a) regularization which avoids artifacts while preserving salient edges. Both TV and TGV regularizations depend on gradient image and can produce false edges at low bit rates. Recently, structure tensor regularization (Prasath et al. 2015, 2019; Prasath and Thanh 2019) is shown to be superior to traditional TV based regularizers in terms feature preservation and image edges restoration. By combining multiple scale computations instead of global gradient regularization functionals these tensor regularization models obtain better accurate results without edge blurring effects.

In this paper, we utilize a multiscale structure tensor edge map driven variational model for JPEG compression artifact removal. By using a geometric neighborhood based structure tensor eigenvalues in a spatially varying, robust edge map we define a variable exponent regularization which adapts to image contents. Image fidelity with color subsampling and DCT operator composition is used as constraint with our regularization based minimization method. By utilizing structure tensor eigenvalues driven edge map to guide the regularization and smoothing process in compressed images, we obtain better edge preservation compared to TV and TGV regularization functionals. Multiscale computation helps retain edges from objects at different scales whereas traditional gradient based regularization methods can blur the edges of objects due to global smoothing properties. Further, compared to JPEG2000 which uses discrete wavelet transform instead of the block-based DCT, our proposed multiscale tensor regularization (MTR) obtained better block based peak signal to noise ratio (PSNR, dB) gains. Experimental results on natural and cartoon/clip-art color images at various compression quality levels show that our proposed method can obtain visibly pleasing resultant images with artifacts reduced considerably. Compared to other methods such as the TV and TGV and other L0 regularizaiton and weighted median filter based models (Xu et al. 2011; Pang et al. 2015; Zhang et al. 2014; Dong et al. 2015) we obtained better performance in terms of blocky artifact reduction and error metrics (Wang et al. 2004; Yim and Bovik 2011).

The paper is organized as follows: Sect. 2 studies the multiscale tensor based regularization for JPEG artifact removal. In Sect. 3 we show the improved performance using our proposed scheme on a variety of compressed images and present comparison results with the literature. Finally, Sect. 4 concludes the paper.

2 Multiscale structure tensor based regularization for artifact removal

2.1 Multiscale structure tensor feature map

Structure tensor captures neighborhood characteristics better than point-wise gradients and is given by

$$J_\rho(\nabla u_\sigma) := G_\rho \star (\nabla u_\sigma \otimes \nabla u_\sigma^T). \tag{1}$$

Here $G_\rho(\mathbf{x}) = (\rho\sqrt{2\pi})^{-1} \exp\left(-|\mathbf{x}|^2/2\rho^2\right)$ is the isotropic Gaussian kernel, $\star$ denotes the convolution operator, $\otimes$ tensor product, $\mathbf{x} \in \Omega$ the image domain, and superscript T is the transpose. The function $u_\sigma := G_\sigma \star u$ encodes the *noise scale* σ and is directly related to the amount of noise present in a given image. Moreover, structures smaller than scale σ are smoothed out and choosing this parameter is important in obtaining better restorations (Prasath and Thanh 2019). In natural images objects occur at different scales, thus a multiscale edge map requires integrating multiple scales (Prasath et al. 2015). For this purpose, we consider adding the gradients in the structure tensor,

$$J_\rho(\nabla u) := \cup_\sigma [G_\rho \star (\nabla u_\sigma \otimes \nabla u_\sigma^T)] \tag{2}$$

Here the range of scales $[\sigma_l, \sigma_L]$ chosen appropriately. We denote corresponding eigenvalues $(\lambda_+^*, \lambda_-^*)$ of the summed up tensor at multiple scales (2). Then, we consider a spatially varying exponent map,

$$p(\mathbf{x}) = 1 + \exp\left[-\frac{2\,\lambda_+^*(\mathbf{x})\,\lambda_-^*(\mathbf{x})}{(k + \lambda_+^*(\mathbf{x}) + \lambda_-^*(\mathbf{x}))}\right] \exp\left[-2\left(\frac{\lambda_+^*(\mathbf{x}) - \lambda_-^*(\mathbf{x})}{k + \lambda_+^*(\mathbf{x}) + \lambda_-^*(\mathbf{x})}\right)^2\right] \tag{3}$$

where a small parameter $k > 0$ is added for numerical stability. This provides edge maps for digital images which are robust to noise than simple gradient based edge detectors from the literature (Prasath et al. 2019). For color images we use the summed up structure tensor formula and compute our multiscale structure tensor and its corresponding eigenvalues.

2.2 Structure recovery under JPEG noise/compression artifacts

The exponent map (3) can be used in a variational energy minimization model,

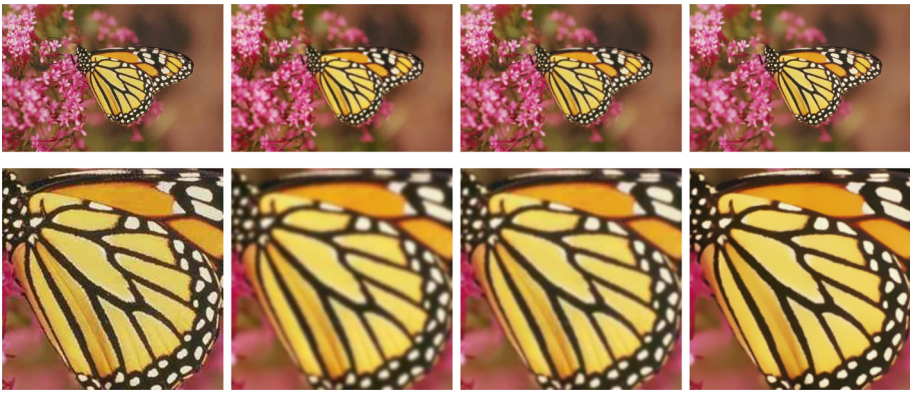
$$\min_u \left\{ E(u) = \lambda \int_\Omega |\nabla u|^{p(\mathbf{x})}\, d\mathbf{x} + F(u(\mathbf{x})) \right\} \tag{4}$$

with $\lambda > 0$ the regularization parameter (Prasath et al. 2018a). The exponent map $p(\cdot)$ need to be constrained to be a smooth monotonically decreasing function such that $\lim_{s\to 0} p(s) = 2$ and $\lim_{s\to\infty} p(s) = 1$. Our $p(\mathbf{x})$ in (3) satisfies these constraints and reflect salient edges in

Algorithm 1 Tensor regularization for JPEG decompression

INPUT: A color JPEG compressed image I with compression.
OUTPUT: Returns a decompressed image u.

1: **procedure** REGULARIZATION
2: Compute the spatial varying exponent map $p(\cdot)$, See Eqn. (3).
3: Compute the image fidelity $F(I(\cdot)))$, See Eqn. (5)
4: Solve the minimization E for u, see Eqn. (4)



(a) Input **(b)** TV (Alter et al, 2005) **(c)** TGV (Bredies and Holler, 2012a) **(d)** Ours

Fig. 1 Our proposed MTR method obtains better result when compared to other regularization methods with data fidelity for JPEG compression artifacts removal. Top row: Artifact removed images, bottom row: zoomed up region details. **a** Input natural color JPEG compressed image with compression quality 30, **b** TV (Alter et al. 2005), **c** TGV (Bredies and Holler 2012a), **d** our proposed MTR method

a given image. Since the DCT based compression artifacts are smaller in scale they are considered as noise and our smoothing technique performs by inhibiting them while preserving large scale edges. Following Bredies and Holler (2012a), we consider the following image fidelity based on the composition of color subsampling and DCT operator composition,

$$F(u(\mathbf{x})) = \begin{cases} 0 & \text{if } u \in U_D, \\ \infty & \text{else,} \end{cases} \tag{5}$$

where the data set is $\{U_D = u \in U | ASu \in D\}$. Here A is the basis transformation operator and in the case of JPEG compression,

$$(Az)_{i,j} = C_i C_j \sum_{k,l} z_{k,l} \cos\left(\frac{\pi(2k+1)i}{16}\right) \cos\left(\frac{\pi(2l+1)j}{16}\right)$$

where $(z_{k,l})_{0\le k,l\le 7}$ are the 8×8 disjoint blocks from each color component. The chroma subsampling operator S is defined 2×2 blocks for each component $(z_{k,l})_{0\le k\le 1}$,

$$Sz = \frac{1}{4} \sum_{k,l=0} z_{k,l}.$$

It is common in DCT based compression that the chroma components Cb, Cr in YCbCr space are reduced by the factor 4 as the human vision is less sensitive in color variations than in the luminance variations. Algorithm 1 provides the pseudo-code of the proposed decompression method.

Figure 1 shows an example artifact reduction on a color image with JPEG compression quality 30 using TV (Alter et al. 2005), TGV (Bredies and Holler 2012a) and our multiscale tensor regularization (MTR) (4). We used the same fidelity term (5) for comparison in all regularization methods, and we can see from the close-up regions that our proposed method obtains better artifact reductions among the regularization models including the original TGV formulation.

3 Experimental results

We utilized the corresponding Euler-Lagrange update equation of the MTR (4) using an implicit finite differences (step size 1/4) (Prasath et al. 2018a), set the regularization parameter $\lambda = 1$ (Prasath et al. 2018b), and it takes approximately 50 iterations to obtain optimal JPEG artifact reductions. Our proposed MTR (4) relies on the number of scales $[\sigma_l, \sigma_L]$ which we chose as [1, 2, 3, 4] and the terminal time for the corresponding dynamic problem is chosen according to maximum performance in block based peak signal to noise ratio (PSNR-B, dB) (Yim and Bovik 2011). We utilized a set of 100 cartoon/clip-art images, and the natural LIVE1 image quality database.[1] We compare our MTR artifact reduction performance with the following methods from the literature:[2] image analogies (IA) (Wang et al. 2006), mean shift (MS) (Comaniciu and Meer 2002), bilateral filter (BLF) (Tomasi and Manduchi 1998), block matching-3D filtering (BM3D) (Dabov et al. 2007),[3] mean-shift (MS) (Comaniciu and Meer 2002), $L0$ gradient minimization (LGM) (Xu et al. 2011),[4] improved LGM (ILGM) (Pang et al. 2015), weighted medial filter (WMF) (Zhang et al. 2014),[5] convolutional neural network (AR-CNN) (Dong et al. 2015),[6] total variation (TV) (Alter et al. 2005), total generalized variation (TGV) (Bredies and Holler 2012a).

In Fig. 2 we compare the performance of different regularization methods on the $Lighthouse3$ image (compression quality 20) from LIVE1 database along with AR-CNN. Compared to TV and TGV regularizations we obtain better artifact reduction and visual quality is comparable to the result obtained by AR-CNN. The PSNR (dB), PSNR-B (dB), and MSSIM (Wang et al. 2004)[7] results indicate our output image removes block based artifacts effectively without color mixing or streaking near edges. Note that PSNR-B is a specifically designed error metric for JPEG compression artifact removal performance in contrast to the general error metrics PSNR, and mean SSIM (Yim and Bovik 2011).

Figure 3 shows an example JPEG artifact removal on a low quality clip-art image (compression quality 10) compared with various edge preserving smoothing/denoising filters (Tomasi and Manduchi 1998; Alter et al. 2005; Dabov et al. 2007; Xu et al. 2011; Bredies and Holler 2012a; Pang et al. 2015; Zhang et al. 2014) as well as segmentation and other methods (Comaniciu and Meer 2002; Wang et al. 2006; Dong et al. 2015). Figure 4 shows average PSNR-B (dB) values for various tested artifact reduction methods on a set of 100 color cartoon/clip-art with JPEG compression quality values ranging from 10 (worst) to 90 (best). As can be seen, our proposed MTR method obtained better results in PSNR-B than other regularization based models for quality > 30. Especially at low bit rates (quality < 30) BLF (Tomasi and Manduchi 1998), BM3D (Dabov et al. 2007), TV (Alter et al. 2005), TGV (Bredies and Holler 2012a), LGM (Xu et al. 2011), ILGM (Pang et al. 2015), WMF (Zhang et al. 2014) perform poorly due to blocky edges being treated as object edges. Table 1 shows the PSNR-B, MSSIM results on 29 LIVE1 natural images with our MTR method outperforming TV, and TGV regularizations. We note that the JPEG2000 uses discrete wavelet transform instead of the block-based DCT utilized in JPEG. Table 2 shows the

[1] http://live.ece.utexas.edu/research/quality/subjective.htm.

[2] ARCNN method was run on the Y component (luminance) of the image converted to the YCbCr color space, and remaining methods were run using corresponding vectorial versions.

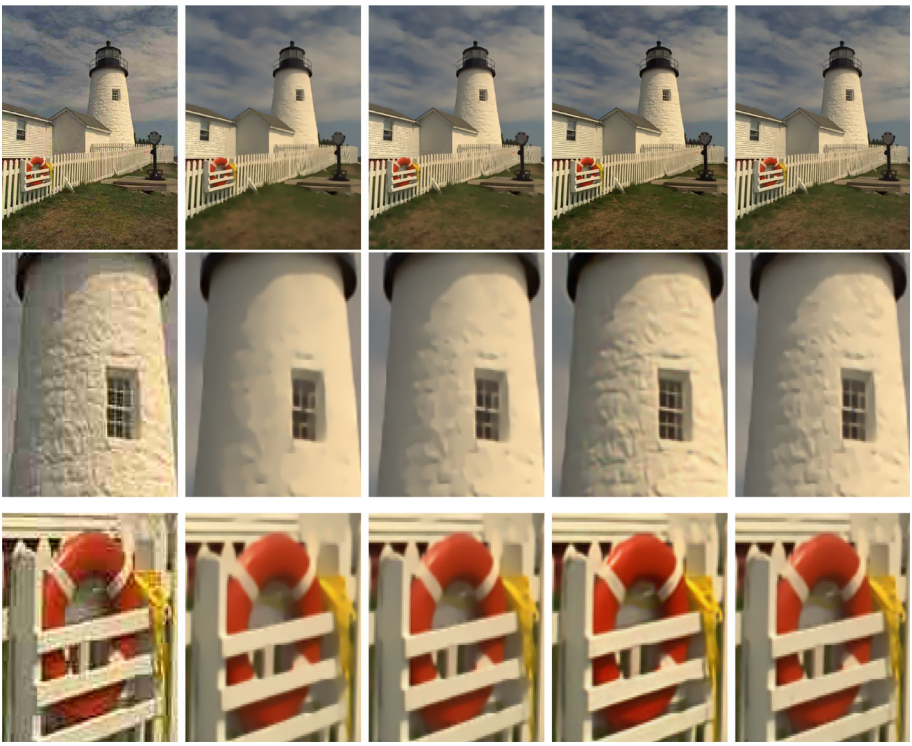
[3] http://www.cs.tut.fi/~foi/GCF-BM3D.

[4] http://www.cse.cuhk.edu.hk/~leojia/projects/L0smoothing/index.html.

[5] http://www.cse.cuhk.edu.hk/leojia/projects/fastwmedian/index.htm.

[6] http://mmlab.ie.cuhk.edu.hk/projects/ARCNN.html.

[7] https://ece.uwaterloo.ca/~z70wang/research/ssim/.



(a) Input **(b)** TV (Alter et al, 2005) **(c)** TGV (Bredies and Holler, 2012a) **(d)** AR-CNN (Dong et al, 2015) **(e)** Ours

Fig. 2 Comparison of various methods on a LIVE1 natural image with JPEG compression quality 20. **a** Input *Lighthouse*3 compressed image, PSNR= 23.31, PSNRB= 22.79, MSSIM= 0.8528, **b** TV (Alter et al. 2005), PSNR= 28.69, PSNRB= 28.58, MSSIM= 0.755, **c** TGV (Bredies and Holler 2012a), PSNR= 29.37, PSNRB= 29.09, MSSIM = 0.8704, **d** AR-CNN (Dong et al. 2015), PSNR= 31.53, PSNRB= 31.31, MSSIM= 0.8495, **e** Our proposed MTR method, PSNR= 29.94, PSNRB= 29.79, MSSIM= 0.7979

PSNR-B, MSSIM results on 5 classical images, namely Baboon, Barbara, Boats, Lena, and Peppers with our MTR method outperforming TV, and TGV regularizations. ARCNN outperforms all methods considered here[8] however, this level of performance required training on the BSDS500 dataset[9] (400 images). In contrast, our method does not require training and relies on robust, multiscale edge map for image regularization for artifact reduction. In terms of the computational complexity, the TV and TGV models require $\sim 2s$ for a color image of size 256×256 whereas our model requires $\sim 3s$ on a MacBook Pro Laptop with 3.8 GHz Intel core i5 processor with 8 Gb RAM. The deep learning model ARCNN requires $\sim 2hr$ for training with a GPU, though the testing on an image only takes $\sim 1s$.

[8] Compare to results from ARCNN (Dong et al. 2015).

[9] http://eecs.berkeley.edu/Research/Projects/CS/vision/grouping/resources.html.

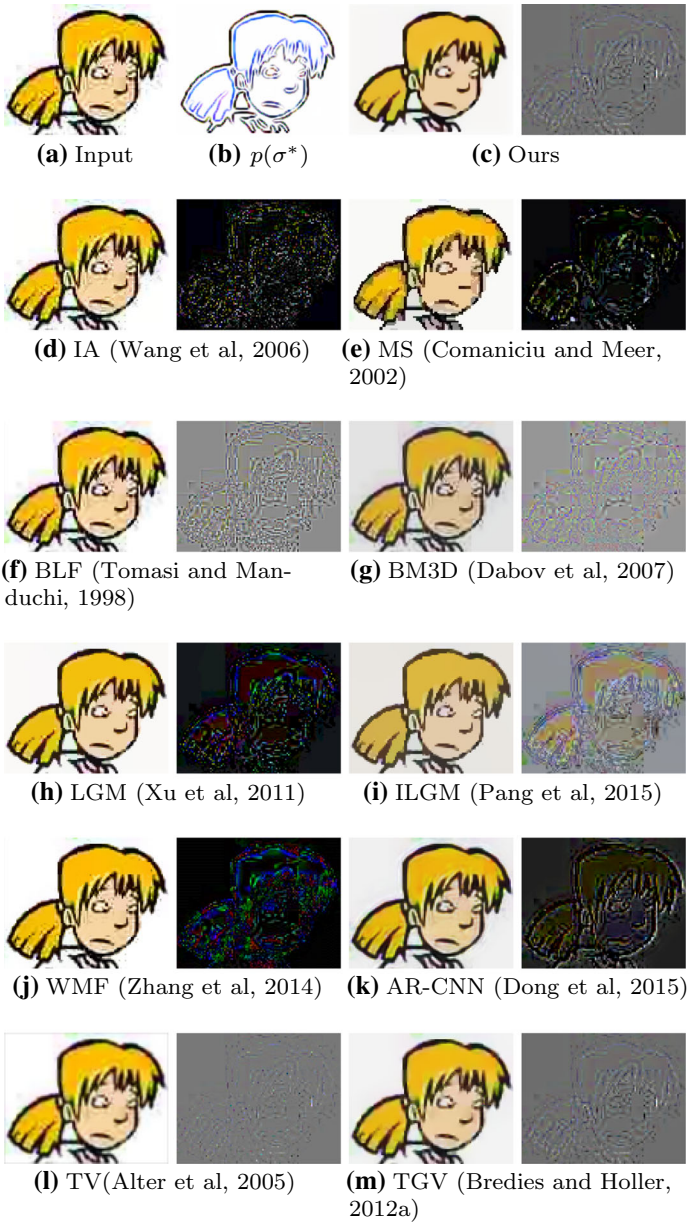


Fig. 3 Our proposed MTR method obtains better result when compared to other methods for cartoon/clip-art JPEG compression artifacts removal. **a** Input JPEG compressed image with low quality (10), PSNR= 10.87, PSNRB= 10.40, MSSIM= 0.7027, **b** multiscale exponent map (3), **c** our MTR result, PSNR= 26.50, PSNRB= 23.17, MSSIM= 0.9248, **d** IA (Wang et al. 2006), PSNR= 13.57, PSNRB= 12.95, MSSIM= 0.5621, **e** MS (Comaniciu and Meer 2002), PSNR= 13.80, PSNRB= 13.22, MSSIM= 0.5871, **f** BLF (Tomasi and Manduchi 1998), PSNR= 30.80, PSNRB= 21.34, MSSIM= 0.9543, **g** BM3D (Dabov et al. 2007), PSNR= 22.95, PSNRB= 21.38, MSSIM= 0.9341, **h** LGM (Xu et al. 2011), PSNR= 13.81, PSNRB= 13.15, MSSIM= 0.6098, **i** ILGM (Pang et al. 2015), PSNR= 24.27, PSNRB= 20.21, MSSIM= 0.8940, **j** WMF (Zhang et al. 2014), PSNR= 24.08, PSNRB= 21.85, MSSIM= 0.9282, **k** AR-CNN (Dong et al. 2015), PSNR= 24.15, PSNRB= 21.54, MSSIM= 0.9661, **l** TV (Alter et al. 2005), PSNR=19.42, PSNRB= 19.34, MSSIM=0.8631, **m** TGV (Bredies and Holler 2012a), PSNR= 32.09, PSNRB= 22.71, MSSIM= 0.9692. In each case, we show the residue image computed as the difference between JPEG compressed and result of artifact reduction techniques

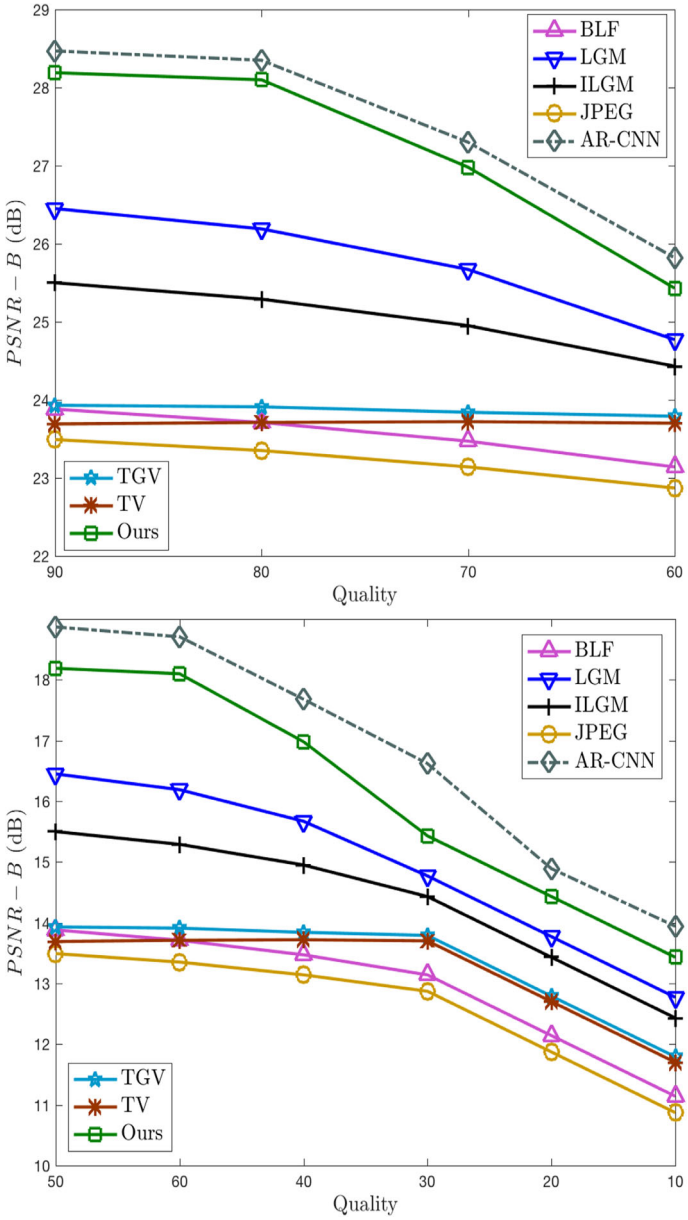


Fig. 4 Average PSNR-B (dB) against JPEG compression quality for different JPEG decompression levels with various smoothing/deblocking methods on a set of 100 cartoon/clip-art color images. Note that x-axis and the y-axis given above are different (Color figure online)

Table 1 The average results of PSNR-B (dB)/MSSIM on the 29 LIVE1 natural images for different quality (Q) levels

Q.	JPEG	JPEG2000	TV (Alter et al. 2005)	TGV (Bredies and Holler 2012a)	Ours	AR-CNN Dong et al. (2015)
10	25.33	26.18	26.35	26.81	*27.45*	**28.74**
	0.7905	0.8132	0.8023	*0.8162*	0.8120	**0.8232**
20	27.57	28.53	28.01	28.59	*29.47*	**30.69**
	0.8683	0.8728	0.8694	0.8703	*0.8816*	**0.8886**
30	28.92	30.19	29.78	30.01	*30.48*	**32.15**
	0.9000	0.9138	0.8276	0.8994	***0.9211***	0.9166
40	29.96	31.50	30.10	31.26	*31.70*	**33.12**
	0.9173	0.9189	0.9143	*0.9215*	0.9207	**0.9306**

Boldface indicate the highest performance among all methods and italics only among regularization methods based on TV, TGV and ours

Table 2 The average results of PSNR-B (dB)/MSSIM on classic-5 test images—Baboon, Barbara, Boats, Lena, Peppers—for different quality (Q) levels

Q.	JPEG	TV (Alter et al. 2005)	TGV (Bredies and Holler 2012a)	Ours	AR-CNN Dong et al. (2015)
10	25.21	26.25	26.43	*27.34*	**28.75**
	0.7800	0.7915	0.8056	***0.8114***	0.8111
20	27.50	27.86	28.41	*29.87*	**30.60**
	0.8541	0.8554	0.8581	*0.8677*	**0.8694**
30	28.94	29.77	29.95	*30.48*	**31.99**
	0.8844	0.8038	0.8172	*0.8531*	**0.8967**
40	29.92	30.05	31.21	*31.85*	**32.80**
	0.9011	0.8969	0.9014	*0.9073*	**0.9101**

Boldface indicate the highest performance among all methods and italics only among regularization methods based on TV, TGV and ours

4 Conclusion

We considered a constrained variable exponent regularization method for JPEG artifact reduction and blocking artifact removal which is based on multiscale structure tensor based feature map. By combining a spatially variable exponent map based smoothing with image fidelity we obtain an effective method to for JPEG deblocking. Our evaluation indicates that our multiscale tensor regularization (MTR) method obtains consistent results on color cartoon, and natural JPEG compressed images. The method is general in the sense that it can be used to obtain artifact reduction in other block DCT-based compressions (Yim and Ik Cho 2005) as well. Extending our approach to handle multichannel imaging (Karaca and Güllü 2019), zooming in JPEG compressed images (Bredies and Holler 2012a) and testing other regularization parameter options (Prasath et al. 2018b) are interesting future works.

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